

The AI Literacy Paradox:

Does Technical Knowledge Undermine Metacognitive Awareness?

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Abstract

This paper starts from a mildly irritating (and, at times, counterintuitive) short circuit: the more technically proficient you become at using generative AI (prompting, parameters, APIs, and so on), the more likely you are to become *less* able to notice when you are wrong (or when you are delegating poorly). In other words, confidence increases, but the ability to monitor the quality of what you are doing does not necessarily increase with it, especially when tasks become genuinely complex. A further complication matters in real settings: most day-to-day AI use (including in companies) is concentrated on low-stakes, low-cognitive-value activities. Rather than training critical thinking and metacognition, this tends to reinforce comfortable routines: they work often enough that you start treating them as if they always work. The next step is predictable: you carry the same confidence (and the same level of vigilance) into domains where errors have a cost.

In the background sits a linguistic and conceptual problem: an anthropomorphic vocabulary that makes machines sound “human”, and, on the other side, a computational vocabulary that makes minds sound “machine-like”. This exchange is not innocent: it creates wrong expectations and epistemic confusion.

If this paper has a practical use, it is in the closing proposal: to prevent AI literacy from turning into a trap, training must go beyond technique and include (1) metacognitive calibration, (2) critical language literacy, and (3) concrete verification and validation habits embedded in everyday workflows.

Contents

1	Introducing the paradox and its linguistic-conceptual roots	3
2	The banality of actual use: when technical competence meets low practical significance	4
3	Systemic manifestations of the paradox: evidence from organizational contexts	5
4	The organizational paradox: systemic overconfidence and the crisis of competence ontologies	6
5	Amplifying the paradox through differential engagement	7
6	Implications for educational and organizational interventions: beyond technical literacy toward metacognitive awareness	8
7	Conclusions: toward epistemic maturity in the age of generative AI (maybe)	9
	References	10

1 Introducing the paradox and its linguistic-conceptual roots

It was inevitable that the short circuit would happen sooner or later. Statistically, it was only a matter of time.

Since the “gold rush” of corporate AI adoption began (a metaphor that fits the current situation: masses of people who can barely tell pyrite from copper suddenly becoming “materials engineers” overnight, drawn by outsized promises while downplaying the risks and difficulty of the “search”, with tangible benefits arriving mainly for those who sell shovels, pickaxes, and sieves), one of the few pillars I have always felt able to defend, with intellectual honesty, as not merely useful but necessary, is training. Making people aware of costs and benefits, and equipping them with a few baseline heuristics so they can at least manage, if not fully understand, the tools they are enthusiastically using has always felt like a moral imperative more than a business exercise.

Driven by the educator’s conviction that knowing is better than not knowing, I threw myself into the quixotic task of convincing corporate decision makers that, if driving a car requires a license earned through weeks of theory and practice, then avoiding damage with a tool engineered to persuade us through plausible syntax without semantics should require at least a couple of days of theory and hands-on practice. And because coherence is the most valuable currency in a trust-based profession, I then became a direct participant in the training journey and, all told, worked with several hundred “corporate students”.

To bring as much added value as possible into these professional exchanges, including what I learn from academic research, I embarked on a relentless program of critical and comparative reading across an enormous volume of papers, studies, and essays (roughly split between serious work and industrial-grade fluff). Over the past weeks, this led me to the present short circuit and to a question I felt compelled to take seriously: could it be that, unlike disciplines where the relationship is broadly one-way (someone understands a “phenomenon”, explains it, I understand it, absorb it, internalize it, and use it), and where study is a precondition for metacognitive awareness, these “disciplines” (or rather, these “phenomena”) establish a relationship that is not one-way, and in practice works differently, perhaps even in reverse?

In fact, during this research, I came across a paper reporting the results of an empirical study by Fernandes et al. [3]. The study describes a counterintuitive phenomenon that challenges traditional assumptions about the link between technical competence and self-evaluation: in the domain of generative AI, individuals with higher AI literacy show lower metacognitive accuracy when they use generative AI systems for complex cognitive tasks. The paradox appears as a critical dissociation between perceived confidence (which increases significantly) and actual ability, not so much to execute efficiently, but to monitor the accuracy of one’s own performance (which remains unchanged or even worsens). The effect is particularly pronounced in the technical-understanding dimension, that is, familiarity with procedural aspects such as prompting (as well as parameter configuration and API usage), which sit at the operational core of interacting with large language models.

To make sense of this, I tried to go beyond individual psychological mechanisms and to question,

at least in part, the epistemological and linguistic foundations on which the contemporary discourse on artificial intelligence has been built. As Floridi and Nobre document [4] in their work on “conceptual borrowing” (a line of argument that is also revisited and expanded in Floridi’s latest book [5]) between AI and the brain and cognitive sciences, there is a systemic linguistic short circuit that precedes and fuels the metacognitive confusions observed among users of AI systems at different levels of expertise.

This short circuit operates in two directions: on the one hand, AI has progressively adopted an anthropomorphic vocabulary that attributes to computational systems cognitive properties that are typically human (attention, learning, memory, reasoning, understanding). On the other hand, cognitive neuroscience has adopted a vocabulary derived from information theory and computer science to describe the brain and mind (encoding, processing, storage, retrieval, decoding).

This reciprocal appropriation is not merely metaphorical; it is systemic, and it generates conceptual confusion with tangible practical consequences, particularly visible in the context of AI literacy and its paradoxical manifestations.

2 The banality of actual use: when technical competence meets low practical significance

As I dug deeper into the systemic manifestations of the paradox in organizational settings, I came across another empirical finding that dramatically amplifies the implications of Fernandes’ results.

An analysis of more than one billion messages sent to ChatGPT between May 2024 and June 2025 [1] reveals a pattern that is both striking and unsettling when set next to the AI literacy paradox: actual use of generative AI tools is concentrated, overwhelmingly, on applications with low cognitive value and limited consequentiality (in the sense of generating meaningful downstream outcomes). Three of the most frequent use-case categories (Practical Guidance, Writing, and Seeking Information) largely include activities such as editing user-provided text, generic advice on a wide range of topics, factual information lookup (a questionable habit, if one truly understands how genAI works), and superficial modifications of existing content. A substantial majority of the conversations classified as Writing do not involve creating genuinely new content, but rather modifying, critiquing, translating, or reformulating text that the user has already produced (for better or for worse).

These patterns point to a paradox within the paradox: if statistics are not merely opinion, even individuals with high technical AI literacy, who therefore display greater confidence but lower metacognitive accuracy when using AI for complex cognitive tasks, still use these tools extensively for elementary operations that require minimal critical evaluation.

The dissociation is deep and systemic. On the one hand, there are individuals with technical competence in prompting, parameter configuration, API usage, and an understanding of how transformers and attention mechanisms work. These elements constitute the core of AI literacy in its procedural dimension.

On the other hand, at least some of these same individuals apply those competencies mainly to trivial tasks, and only a very small fraction (statistically negligible) of messages involve sophisticated work such as software programming, mathematical reasoning, or data analysis. These are precisely the domains where evaluating outputs demands substantial expertise and where the risks associated with unnoticed errors can be material.

The situation becomes even more problematic once we consider that this usage configuration (high technical competence applied to trivial tasks) might paradoxically reinforce, rather than mitigate, the metacognitive deficits hypothesized and empirically “demonstrated” by the research.

Repeated use of AI for low-consequentiality tasks creates ideal conditions for what we might call a degraded habitual metacognition: users develop smooth interaction routines that produce frequently adequate (or merely plausible) outputs on simple tasks. This reinforces global confidence in one’s ability to use AI, without developing the critical-monitoring skills required for genuinely complex work (while also feeding the broader entropy driver of conceptual banality).

When these same users later rely on AI for more complex or potentially “risky” tasks, they behave with an overconfidence calibrated on success in trivial contexts, and they apply it uncritically to situations that would require a significantly higher level of metacognitive vigilance.

Organizational adoption data from the Wharton AI Adoption Report [2] completes the picture with complementary evidence. Nearly half of business leaders use generative AI daily (a significant increase versus the prior year), and the vast majority use it at least weekly. Yet the dominant use cases remain concentrated in areas of individual, procedural productivity, which again fall within the prevailing usage patterns described above. As a result, this “prevailing mode of use” extends beyond personal practice and also penetrates workplaces, where the consequences of underappreciated errors (once these tools are used for more meaningful tasks) can be substantial.

A particularly insidious configuration emerges: individuals (in personal use) and organizations (that is, individuals using AI in professional contexts) develop technical fluency and confidence through mostly trivial applications. They consolidate interaction patterns that minimize critical monitoring, and then they extend those same patterns, and the associated confidence, to domains where a lack of metacognitive accuracy can produce material consequences.

The AI literacy paradox therefore highlights an individual cognitive vulnerability in isolated situations that becomes a systemic pattern reinforced by the very structure of technology adoption: the gateway to generative AI runs through trivial use cases that mask the need to develop metacognitive skills, creating a false sense of mastery that is then transferred inappropriately to contexts that are not consequence-free.

3 Systemic manifestations of the paradox: evidence from organizational contexts

While Fernandes and colleagues document the AI literacy paradox at the level of individual cognitive processes, and the analysis of real-world use cases reveals the prevailing banality of concrete

corporate applications, large-scale survey data from one of the global recruiting players, covering several thousand workers and organizational leaders across twenty-four markets, increasingly suggests that the phenomenon is not confined to the individual psychological dimension. It also manifests as a systemic vulnerability at organizational and societal levels.

The most interesting signal, in my view, concerns the dissociation between self-perceived AI competence and the amount (and quality) of training actually received: while a substantial share of workers claim to believe that AI skills will improve their career and promotion prospects, only a very small minority report having received any form of AI training in the last year. This dramatic gap between expectations and preparation is not merely a matter of unmet demand for training. It is a collective-level manifestation of the paradox itself.

People express confidence in their ability to navigate an AI-dominated working future despite the absence of formal preparation. This suggests that the overconfidence documented at the individual level by Fernandes and colleagues, which in this case is grounded in “unaware” usage experience as a source of literacy, replicates at a macro-social scale. Superficial mastery of consumer AI tools is evidently being confused with genuine competence that would effectively prepare for professional AI integration, exactly as one would expect from the mechanism of the illusion of explanatory depth operating at the individual level.

The systemic nature of the paradox becomes particularly clear once we consider the uneven distribution of training opportunities and self-reported confidence across demographic groups. Here, metacognitive biases and self-confidence patterns replicate what we have seen for decades in terms of gender and generations (higher male overconfidence, greater resistance in Baby Boomers compared to Gen Z, and so on).

4 The organizational paradox: systemic overconfidence and the crisis of competence ontologies

If we extend the analysis from the individual level to organizational decision making, we see the emergence of a form of systemic overconfidence that reproduces, at a macro level, the same metacognitive-deficit mechanisms observed at the individual level (confirming the old adage that organizations are, ultimately, made of people).

An overwhelming share of business leaders agree that adopting AI technologies has improved attraction, engagement, and retention of talent, and they express enthusiasm about the proliferation of AI, automation, and robotics in the workplace. However, only a very small minority of organizations report having built a systematic and structured mapping (an ontology, in practice) of the competencies that are expected, transformed, and in the process of being transformed.

Organizations appear to operate under the implicit assumption that adopting AI technologies will be automatically beneficial and transformative, where “adoption” often means purchasing a handful of licenses for the chatbot deemed most suitable, sometimes accompanied by a light dusting of training (occasionally just a few asynchronous videos with the depth of a TikTok).

This ignores the fact that realizing these benefits requires a deep and structured understanding of which specific competencies are needed, how those competencies relate to AI technologies, and how they must be systematically identified, developed, and validated.

Building a skills ontology is not merely a “nice to have” for HR administration. It is the epistemological infrastructure that allows an organization to have metacognitive awareness of its actual capabilities with respect to AI, just as metacognitive calibration allows individuals to accurately recognize their own limits and gaps. It is a form of herd immunity against AI-at-all-costs shallowness. An organization that proclaims AI-driven transformation without building a skills ontology is the systemic equivalent of an individual with high technical AI literacy but low metacognitive accuracy: operationally efficient, but epistemically confused about the nature and limits of its own capabilities.

5 Amplifying the paradox through differential engagement

A further layer of complexity emerges when we consider the AI literacy paradox within patterns of differential engagement with AI technologies.

Empirical research identifies significant disparities in AI usage across different categories of workers. Knowledge workers use AI at significantly higher rates than frontline workers. Similarly, workers with university or postgraduate education use AI for work-related activities far more frequently than workers with only secondary education. These patterns suggest that exposure to AI, and therefore the probability of developing both technical fluency and potential metacognitive deficits, may be distributed very unevenly across the workforce, with significant implications for equity and inclusion in the AI era.

Particularly concerning is the phenomenon of “shadow” AI use, which raises critical questions about governance, accountability, and organizational risk. If individuals with higher technical AI literacy are also more vulnerable to metacognitive deficits, then widespread, unsupervised AI use by technically capable but potentially overconfident knowledge workers represents a meaningful systemic risk.

It may sound like a social-media topic, but the fear of replacement is real (and mainstream as a narrative). As a result, FOMO (fear of missing out) can generate paradoxical outcomes. On the one hand, there is “AI shame” (culturally, perhaps more present in our Italian organizations than elsewhere), which leads people to deny use even when it is obvious. On the other hand, it can create perverse incentives toward overconfidence: workers may feel compelled to project security and competence in AI usage even when their actual understanding is superficial, in order to avoid appearing obsolete or inadequate.

6 Implications for educational and organizational interventions: beyond technical literacy toward metacognitive awareness

As a first partial answer to my irritating question, I can now start to say that traditional approaches to AI training may be not only insufficient, but potentially counterproductive, unless they are paired with interventions explicitly aimed at metacognitive calibration and at a critical awareness of technical vocabulary.

The practical implication I draw, and that I intend to implement in my educational models on the topic, is to explicitly incorporate three components:

- First, explicit metacognitive training that teaches learners not only how to use AI, but also how to monitor and critically evaluate the quality of their own usage process and the outputs produced. This can be supported by structured self- and peer-explanation practices, metacognitive reflection prompts embedded in workflows, and calibration exercises that help individuals develop a more accurate understanding of their limits and of the situations where external or third-party verification is necessary.
- Second, critical language training that makes students aware of the anthropomorphic vocabulary that permeates the AI field and of the conceptual confusions this vocabulary can generate. Participants in AI training should be explicitly taught the difference between computational and human-cognitive senses of terms such as “attention”, “understanding”, and “reasoning”, and they should develop the habit of systematically disambiguating between these senses when discussing AI capabilities.
- Third, validation-oriented training that teaches practical verification, testing, and quality-assurance skills appropriate to different types of AI outputs and application contexts.

At the organizational level, the implications of the paradox further strengthen the need to fundamentally rethink how organizations govern, monitor, and support AI usage by their workers. There is an urgent need to create environments in which AI use can be transparent, supported, and appropriately supervised without creating incentives for “shadow” usage. This requires a delicate balance: on the one hand, organizations must implement oversight mechanisms that prevent risky or dysfunctional uses of AI; on the other hand, they must avoid overly restrictive policies that would push usage even further into the shadows.

Given what cognitive psychodynamics has taught us for years, namely that consenting adults learn primarily through peer confrontation, the most viable approach is, in my view, the development of organizational “AI champions”. Individuals with specialized training not only in the technical use of AI, but also in principles of metacognitive calibration, critical validation, and linguistic awareness. These champions should provide on-demand support to colleagues who use AI for high-impact tasks, helping them identify high-risk situations where external verification is essential, suggesting appropriate validation strategies, and facilitating critical reflection on their own usage processes.

In parallel, organizations could implement “metacognitive infrastructures” in their internal AI portals (tools that are increasingly common for collecting and sharing internal knowledge and local best practices) and within digital workflows, by incorporating structured prompts that encourage users to reflect critically on the quality and reliability of AI outputs before making decisions based on them.

7 Conclusions: toward epistemic maturity in the age of generative AI (maybe)

We are in the middle of an epistemic transition, driven by generative AI, and characterized by a fundamental tension between the speed at which we evolve our technical capacity to build and use increasingly sophisticated systems and the slowness with which our (personal and organizational) conceptual, epistemological, and metacognitive understanding adapts to this new technological reality.

We are living through a historical moment in which the boundaries between human cognition and artificial computation are becoming progressively blurred in the phenomenological and linguistic sense of everyday experience with these systems. Yet the ontological differences remain deep, and this blurring creates ideal conditions for confusion that can have meaningful practical consequences.

Only through a coordinated, multidisciplinary effort that explicitly addresses this epistemic transition in its cognitive, linguistic, social, organizational, and technological dimensions, while refusing the marketing simplifications of “off-the-shelf technology”, will it be possible to turn an empirically demonstrated vulnerability into an opportunity to develop mature, reflective, and critically informed forms of human expertise in the age of generative artificial intelligence.

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